

Unit 402 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Cable**, 38.9 ac, BAF 20 variable-radius cruise on a 300 ft grid, 19 plots, 203 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	214 sq ft/ac (1 SE of mean, BA 3.1 %)	Type	Sierran mixed conifer
Trees per acre	64	CWHR size / density	5 / D (cover 90 %)
QMD	24.8 in	SDI	261 (45 % of max 576)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	214 sq ft/ac 24.8 in	BA QMD	0 sq ft/ac 0.0 in
TPA	64	TPA	0
Volume	40.6 MBF/ac (8094 cu ft)	Volume	0.0 green tons/ac (0 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~165 sq ft/ac BA	Sampling error (volume)	14.4 % at 95 % (CV 30 %, t = 2.10, n = 19)
Species (BA)	PP 67, WF 65, DF 36, IC 27, SP 18	Stratum standard	40 % of volume - MEETS; plots for standard: 3
		Sampling error (basal area)	6.6 % at 95 % (CV 14 %); secondary, not the standard
		Plot count	19 vs 20-plot local practice, not a handbook standard: SHORT by 1

Cubic volume = total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium) net of defect; Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006); green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber, so applied to total-stem CVTS it overstates sawlog volume; merchantable cubic volume would be the consistent pairing. Sampling error at 95 % confidence with Student's t on plot net cubic volume per acre, the quantity the handbook standards apply to (FSH 2409.12 ch. 40, 41.1(5)(a) and (b)); stratum standard 40 % of volume for tree-measurement sales; the basal-area error is secondary. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables exclude records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
12-15.9	14	13.1	14.7	360	1.8
16-19.9	30	17.5	31.6	872	4.4
20-23.9	33	13.0	34.7	1056	5.3
24-27.9	28	8.0	29.5	1017	5.1
28-31.9	17	3.6	17.9	659	3.3
32-35.9	12	2.0	12.6	534	2.7
36-39.9	20	2.7	21.1	900	4.5
40-43.9	10	1.1	10.5	500	2.5
44-47.9	15	1.4	15.8	811	4.1
48-51.9	5	0.4	5.3	269	1.3
52-55.9	3	0.2	3.2	171	0.9
56-59.9	4	0.2	4.2	241	1.2
60-63.9	12	0.6	12.6	706	3.5
All	203	63.8	213.7	8094	40.6

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	64	16.8	67.4	27.2	2968	14.9
WF	62	20.7	65.3	24.1	2377	11.9
DF	34	12.0	35.8	23.4	1254	6.3
IC	26	9.5	27.4	23.0	818	4.1
SP	17	4.8	17.9	26.1	677	3.4
All	203	63.8	213.7	24.8	8094	40.6

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 402: plots on the 300 ft grid (orange = QA flag)

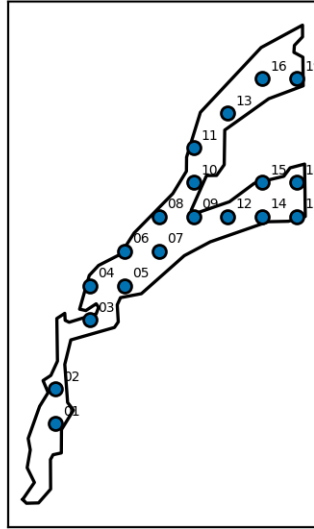


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted; 1 more plots would meet the 20-plot local practice, not a handbook standard

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
402-01	7	140	37	ok
402-02	12	240	53	ok
402-03	10	200	44	ok
402-04	12	240	61	ok
402-05	12	240	83	ok
402-06	9	180	36	ok
402-07	11	220	48	ok
402-08	10	200	56	ok
402-09	10	200	34	ok
402-10	12	240	58	ok
402-11	9	180	39	ok
402-12	11	220	76	ok
402-13	10	200	61	ok
402-14	11	220	62	ok
402-15	10	200	42	ok
402-16	12	240	57	ok
402-17	12	240	46	ok
402-18	13	260	51	ok
402-19	10	200	57	ok